FAP 16S Raw Data Processing Notes - Batches 2022 & 2024

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Raw data description

Raw 16S data

Overall, 166 FASTQ files were received, including forward and reverse reads of 185 samples.

The 16S region amplified is the V3-V4, using primers:
Forward: CCTACGGGNGGCWGCAG
Reverse: GACTACHVGGGTATCTAATCC

The reads are paired-end, 250 nt in each direction.

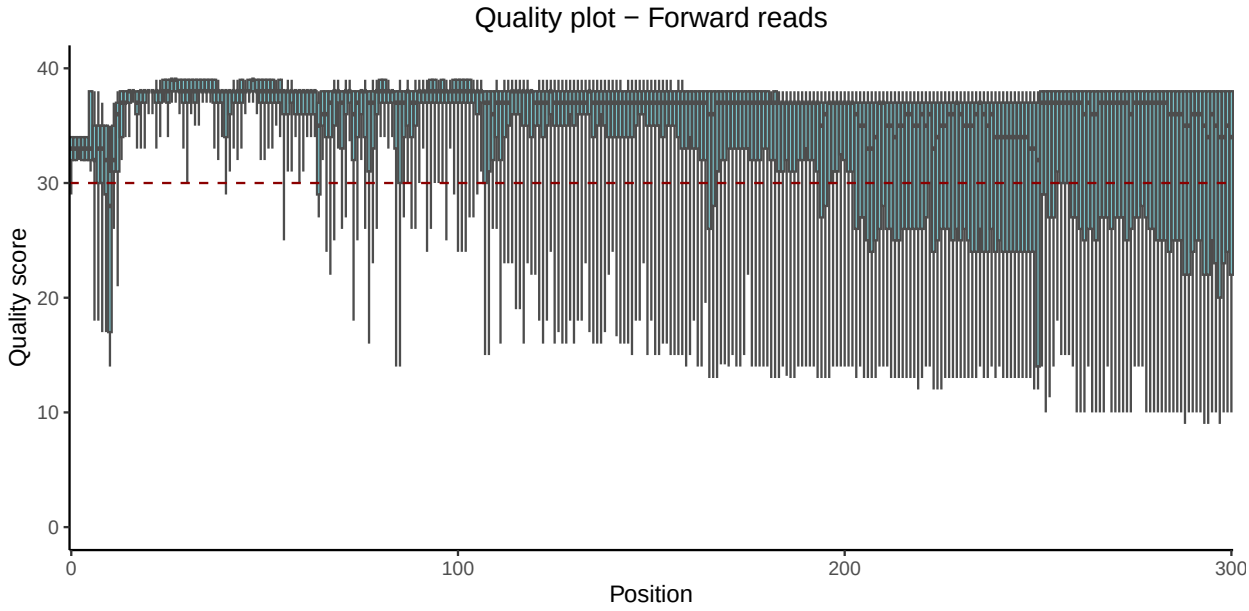
Metadata

TODO - DESCRIBE METADATA

Raw data quality

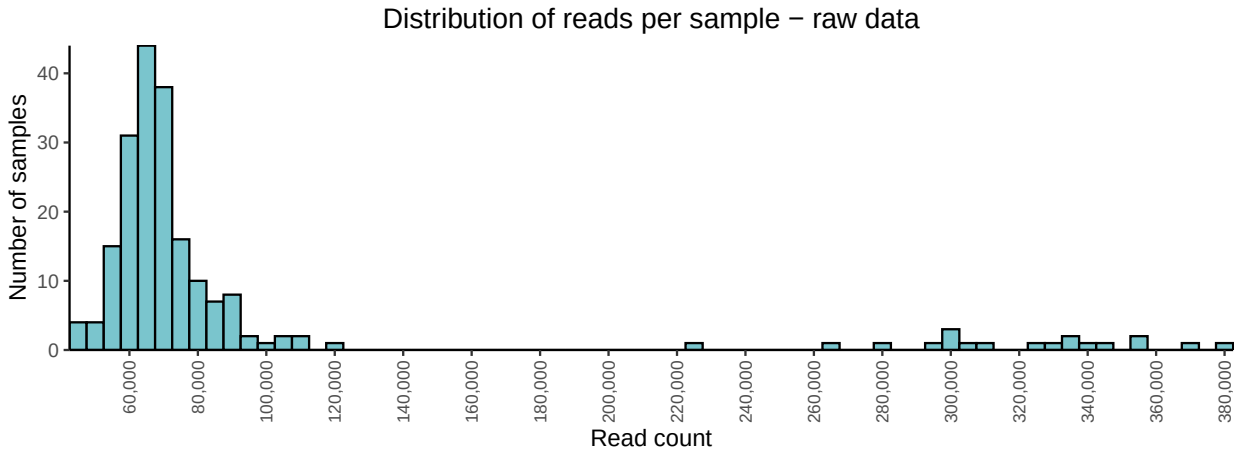
Read quality

The following plots show the distribution of read quality at each position along the read. As a rule of thumb, quality scores over 30 (red dashed line) are considered good.





Sequencing depth



DADA2 denoising

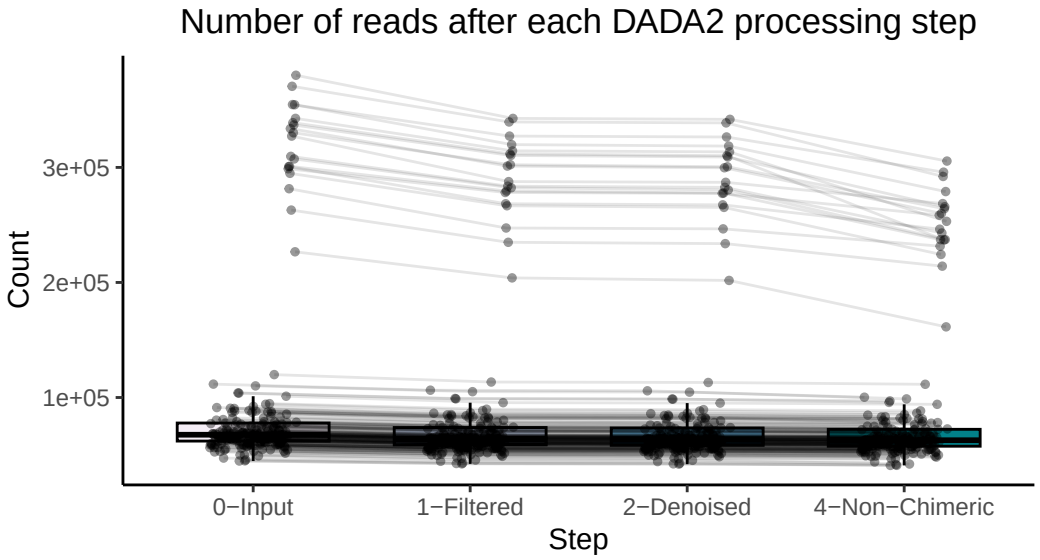
DADA2 algorithm was used for the denoising of raw reads. The DADA2 pipeline specifically includes the following steps:

- Trimming and quality filtering
- Learning error models and finding unique amplicon sequence variants (ASV's)
- Merging paired reads
- Removing chimeras

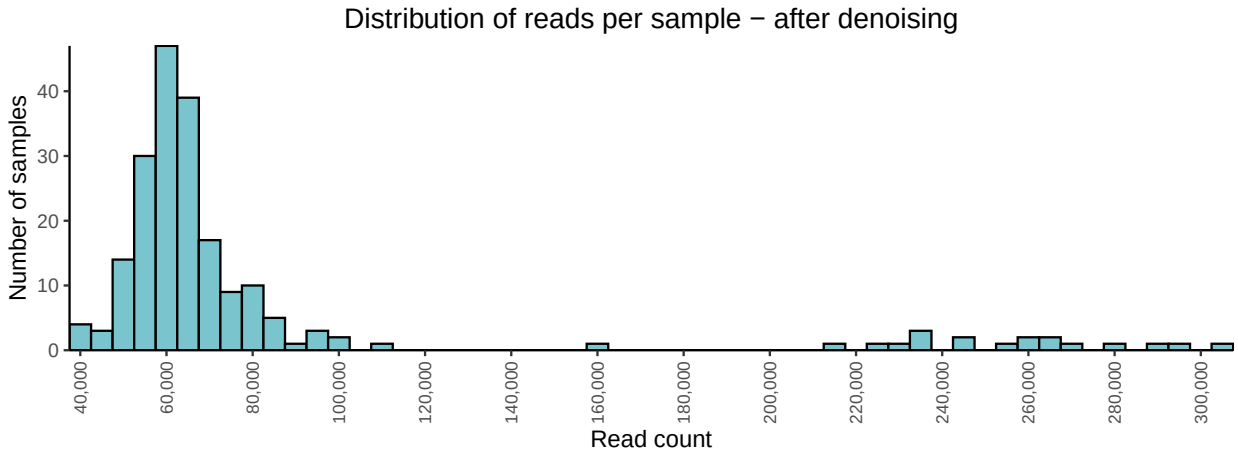
DADA2 processing was performed using the qiime plugin, with the following non-default parameters:

Parameter	Description
`DADA_TRIM_LEFT_F=15`	Forward read - left trimming
`DADA_TRUNC_LEN_F=165`	Forward read - truncation length (0 means no truncation)
`DADA_TRIM_LEFT_R=0`	Reverse read - left trimming (not used)
`DADA_TRUNC_LEN_R=0`	Reverse read - truncation length (not used)
`DADA_MAX_EE_F=3`	Controls the quality filtering done by DADA. Default: 2
`DADA_MAX_EE_R=4`	Same as above, for reverse reads
`DADA_POOLING=pseudo`	Pool samples to identify variants more sensitively.

The following plot provides an overview of the number of reads discarded at each step.



Final distribution of read counts per sample:

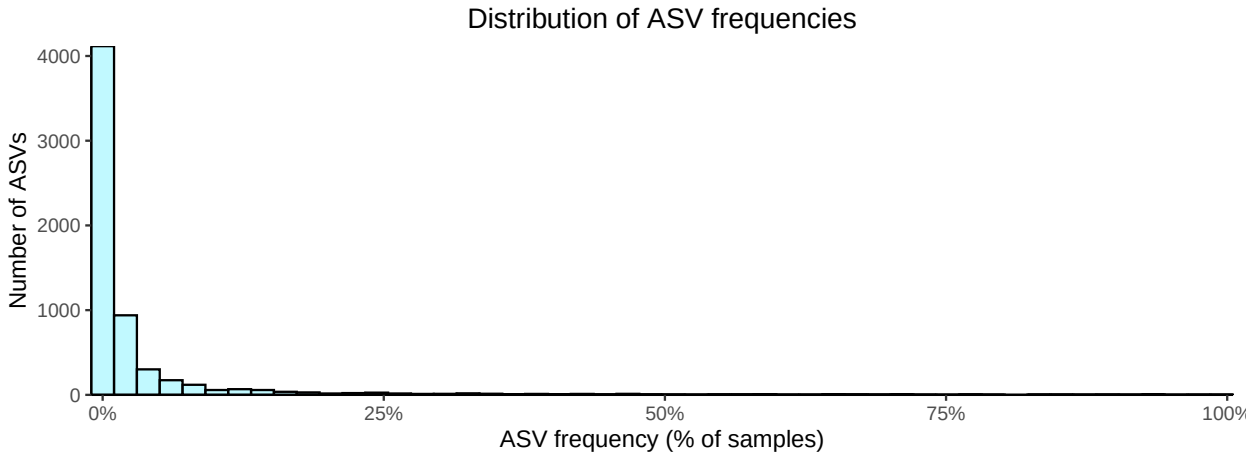


Low-quality samples TODO - EDIT WHEN YOU KNOW THE QUALITY OF THE OTHER BATCH Because of the overall high quality and our aim to keep as many samples as possible, we decided to keep all samples in the analysis. However, it is important to note that some samples have a lower number of reads than others. This may affect the downstream analysis, especially when comparing samples with different read counts.

Identified ASV's

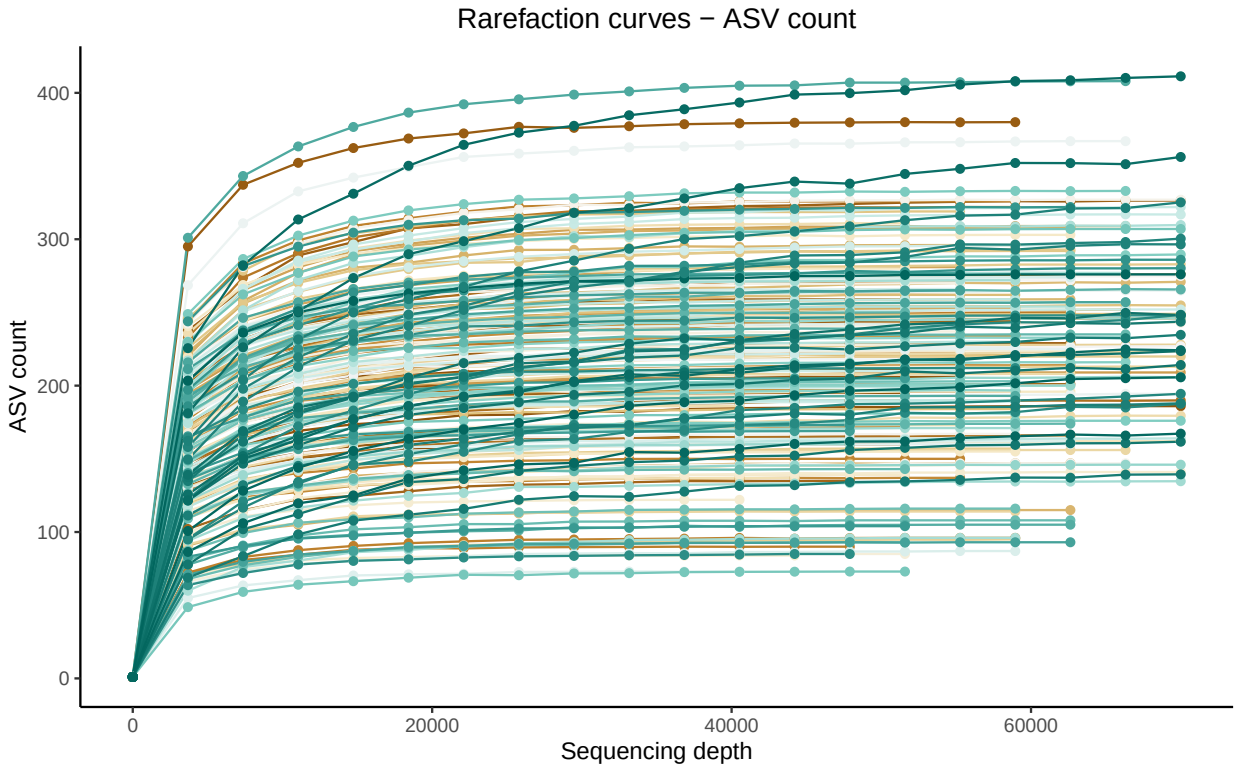
Overall, 6154 unique ASV's were identified. 5678 of which are relatively rare and appear in <10% of the samples.

ASV prevalence (i.e. in how many samples did each ASV appear) are plotted below:



Rarefaction curves

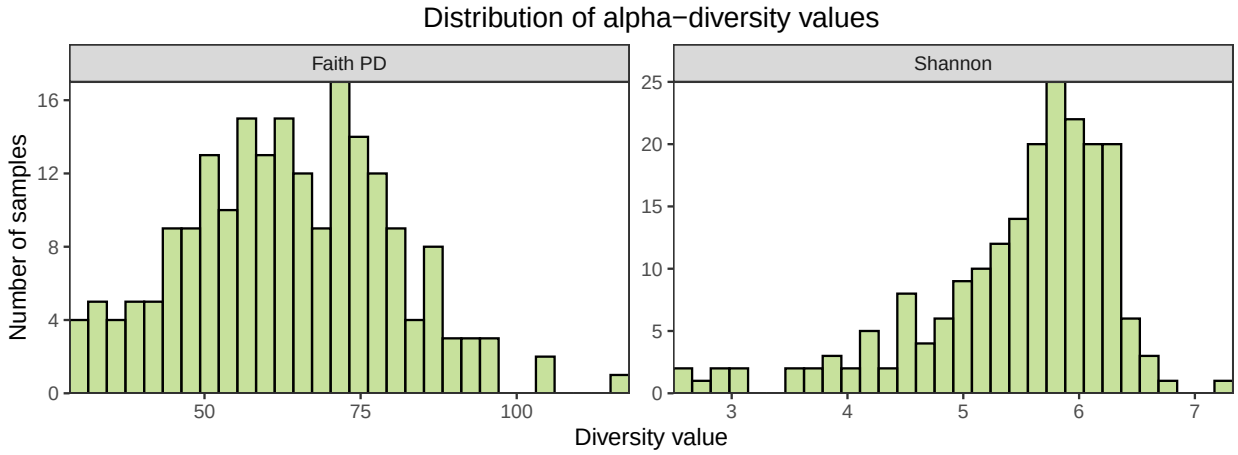
Rarefaction curves describe the number of observed species (here: sequence variants, ASVs) against the number of reads per sample. The curve is created by randomly re-sampling the pool of reads available for each sample and then plotting the average number of ASVs identified from the chosen reads. Generally, it initially grows rapidly and then slightly flattens (as the rarest species remain to be sampled).



Alpha-diversity

Alpha-diversity metrics measure the diversity of the microbial community (at the ASV level) within each sample. The following metrics were computed:

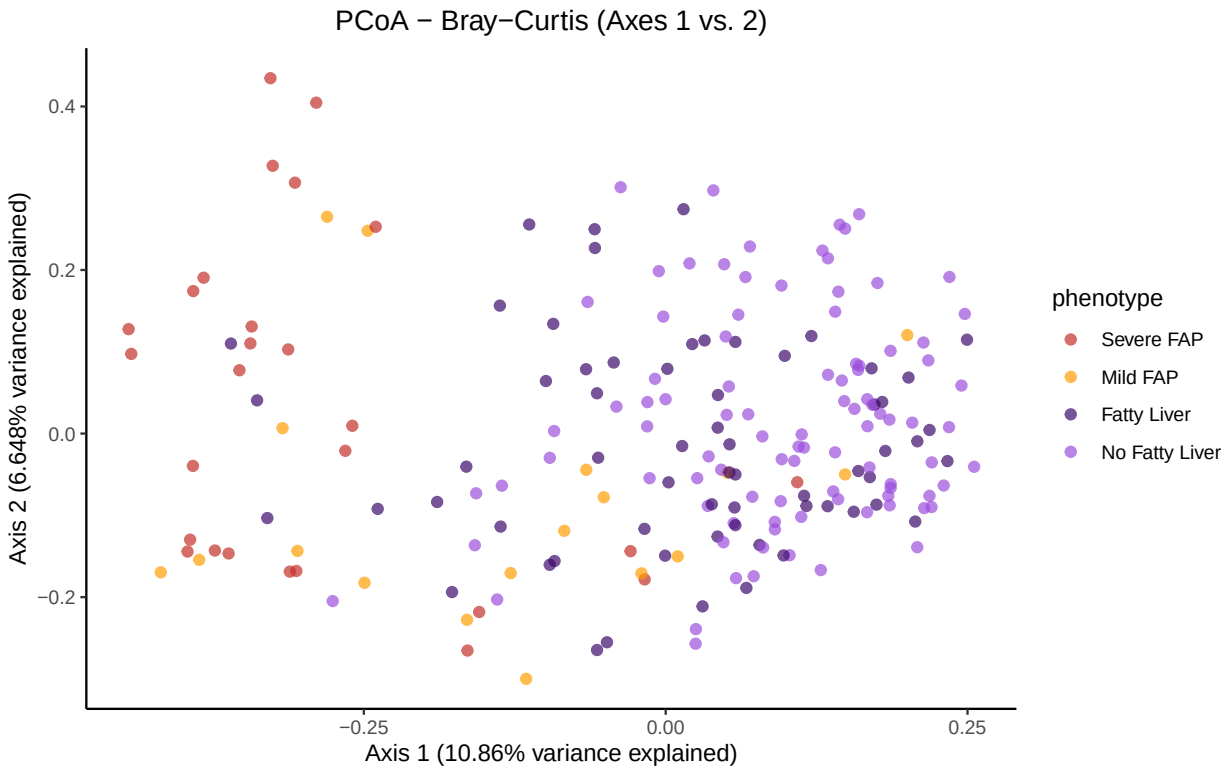
- *Shannon diversity*: This index quantifies the uncertainty in predicting the species identity of an individual that is taken at random from the dataset. The higher the value, the higher the diversity of species is in the particular community. A value of 0 indicates a community that only has one species.
- *Faith phylogenetic diversity*: This index also takes into account phylogenetic relatedness of species within a sample. See here. Higher values indicate a larger spread over the phylogenetic tree, i.e. higher diversity.

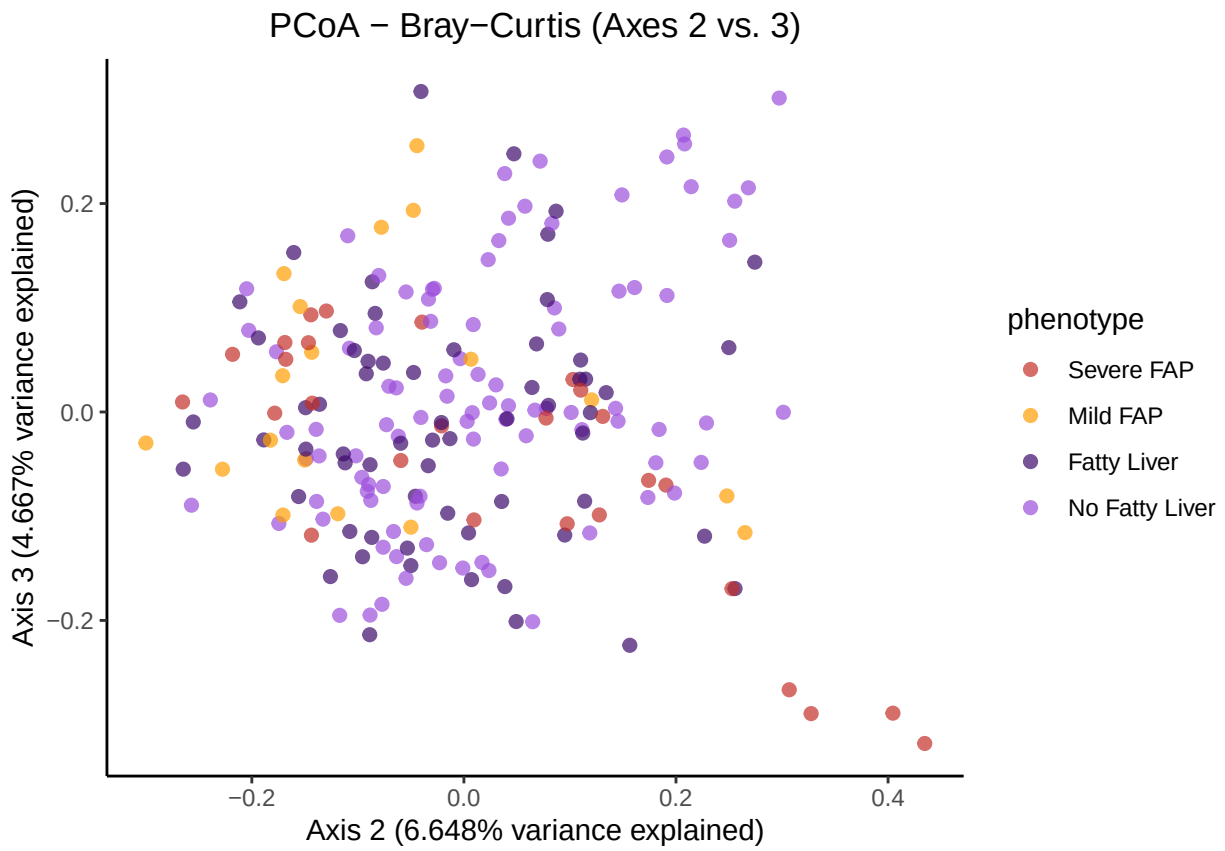


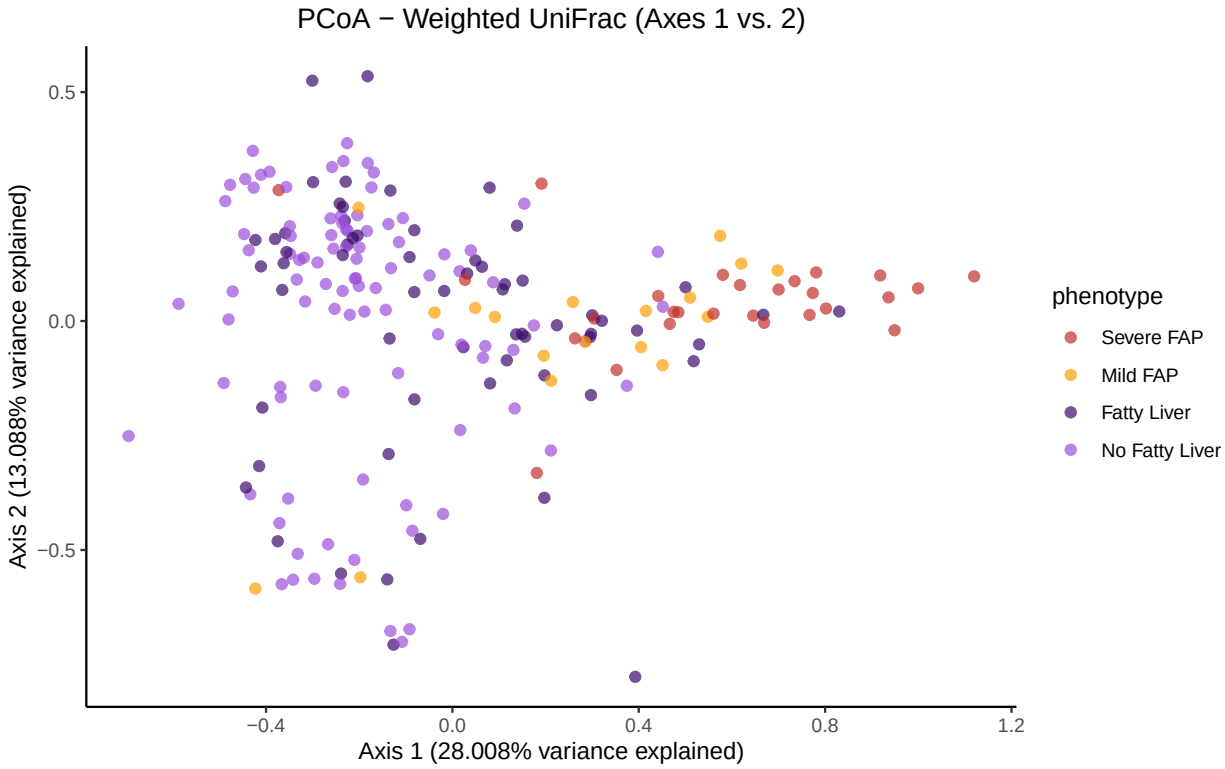
Prinicipal coordinate analysis (PCoA)

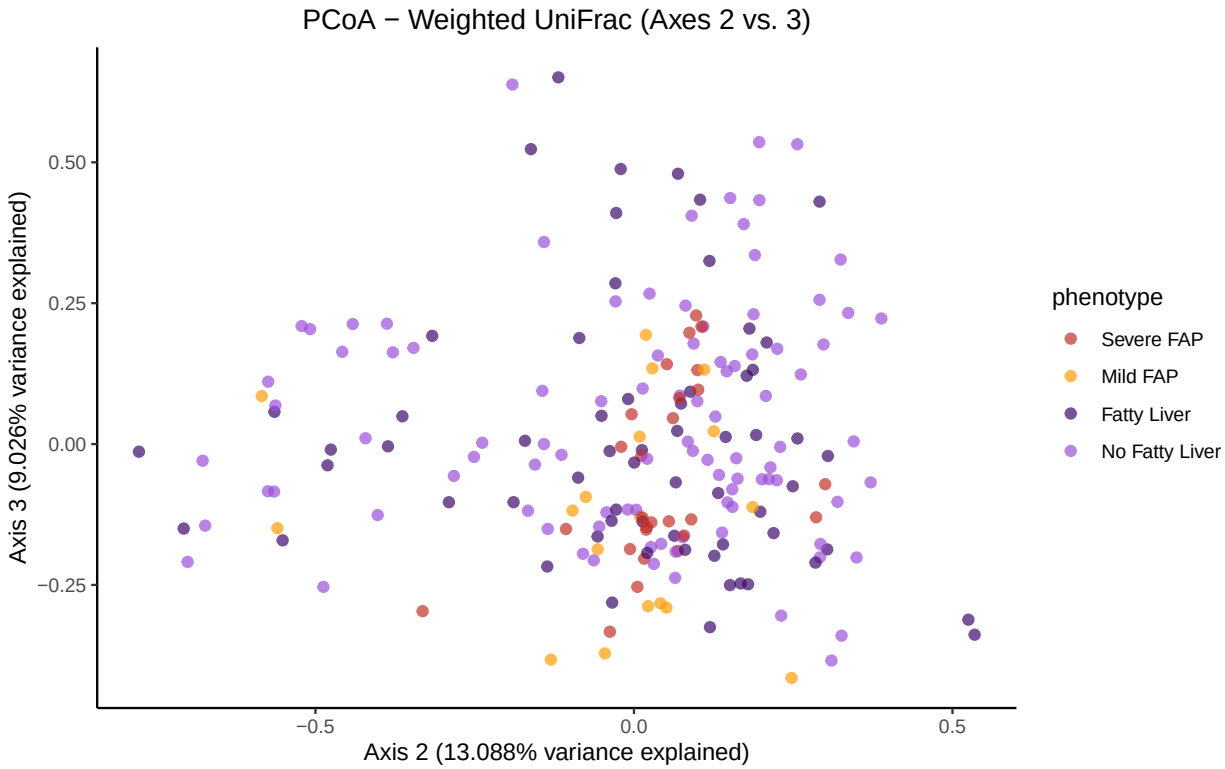
We run PCoA with Bray-Curtis/ Weighted UniFrac distances to show an overview of samples and similarity between them. Samples from the same subject are connected with a line.

```
## Warning: ‘aes_string()‘ was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with ‘aes()‘.
## i See also ‘vignette("ggplot2-in-packages")‘ for more information.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```


PCoA − Bray−Curtis (Axes 1 vs. 2)
Axis 2 (6.648% variance explained)
0.4
0.2
0.0
−0.2
−0.25
0.00
0.25
Axis 1 (10.86% variance explained)
phenotype
Severe FAP
Mild FAP
Fatty Liver
No Fatty Liver


PCoA − Bray−Curtis (Axes 2 vs. 3)
Axis 3 (4.667% variance explained)
0.2
0.0
−0.2
−0.2
0.0
0.2
0.4
Axis 2 (6.648% variance explained)
phenotype
Severe FAP
Mild FAP
Fatty Liver
No Fatty Liver


PCoA − Weighted UniFrac (Axes 1 vs. 2)
Axis 2 (13.088% variance explained)
0.5
0.0
−0.5
−0.4
0.0
0.4
0.8
1.2
Axis 1 (28.008% variance explained)
phenotype
Severe FAP
Mild FAP
Fatty Liver
No Fatty Liver


PCoA − Weighted UniFrac (Axes 2 vs. 3)
Axis 3 (9.026% variance explained)
0.50
0.25
0.00
−0.25
−0.5
0.0
0.5
Axis 2 (13.088% variance explained)
phenotype
Severe FAP
Mild FAP
Fatty Liver
No Fatty Liver

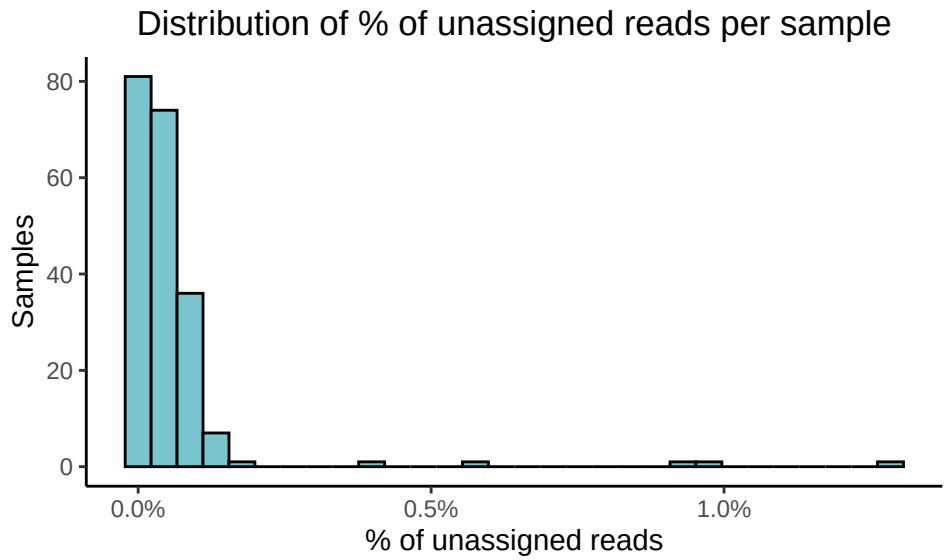
Taxonomy assignment

To assign taxonomy per ASV we used qiime's `feature-classifier` plugin, with SILVA 138 as the reference database. Briefly, we first extracted reads corresponding to the below primers from the reference database, then we train a naive-bayes classifier (using `fit-classifier-naive-bayes`), and lastly we assigned a taxonomy to each ASV in our data using the trained model. The following non-default parameters were used in this process:

Parameter	Description
`F_PRIMER=CCTACGGGNGGCWGCAG`	Forward primer used to extract V3-V4 reads
`R_PRIMER=GACTACHVGGGTATCTAATCC`	Reverse primer used to extract V3-V4 reads
`MIN_AMP_LEN=300`	Remove extracted sequences shorter than this length
`MAX_AMP_LEN=600`	Remove extracted sequences longer than this length

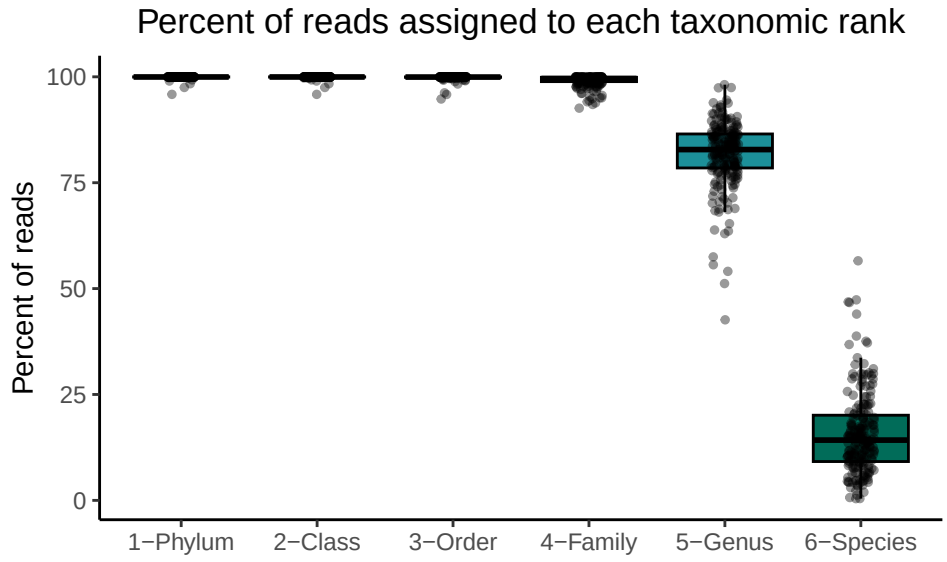
Unassigned reads

Here we calculate the % of reads per sample for which taxonomy was not assigned, i.e. no match was found in the reference database. We expect these percentages to be low.



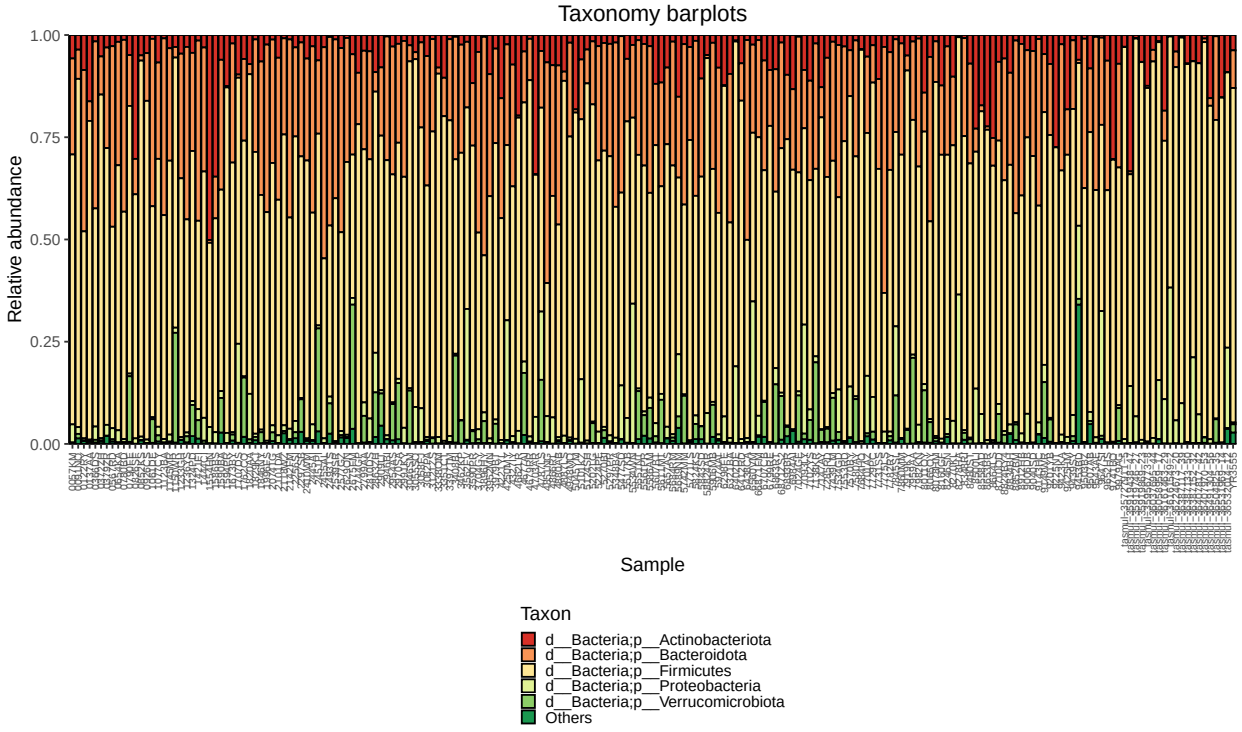
Similarly, here we show the percent of reads succesfully assigned to each taxonomic rank. We note that 16S is considered inefficient in identifying bacteria at the species level or below.

Each dot corresponds to a single sample.

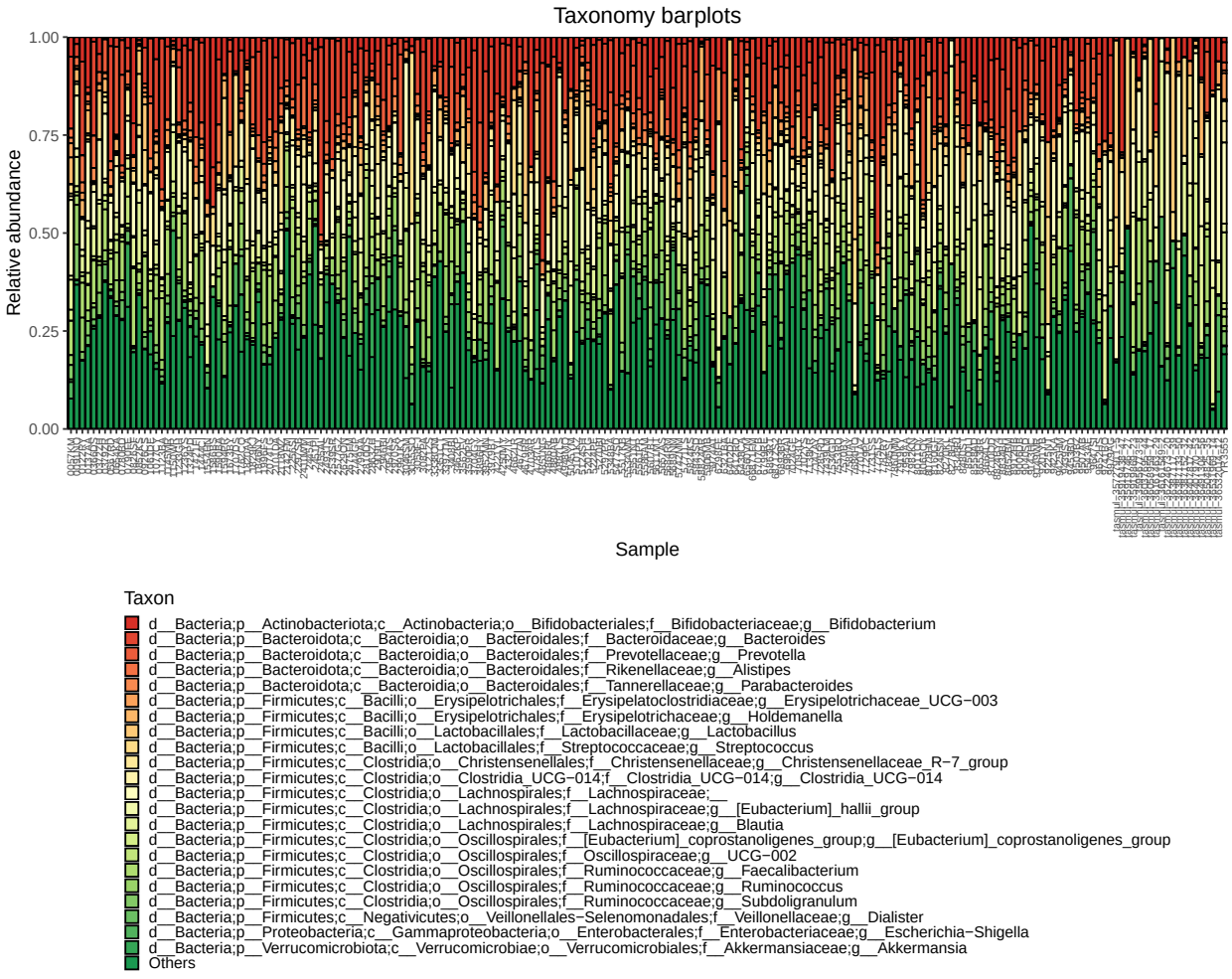


Phylum-level taxonomy barplots

Barplots below describe the phyla (or genera) identified in each sample. Rare taxa were grouped together (using a cutoff of mean relative abundance < 0.01) and are collectively named 'Others'.

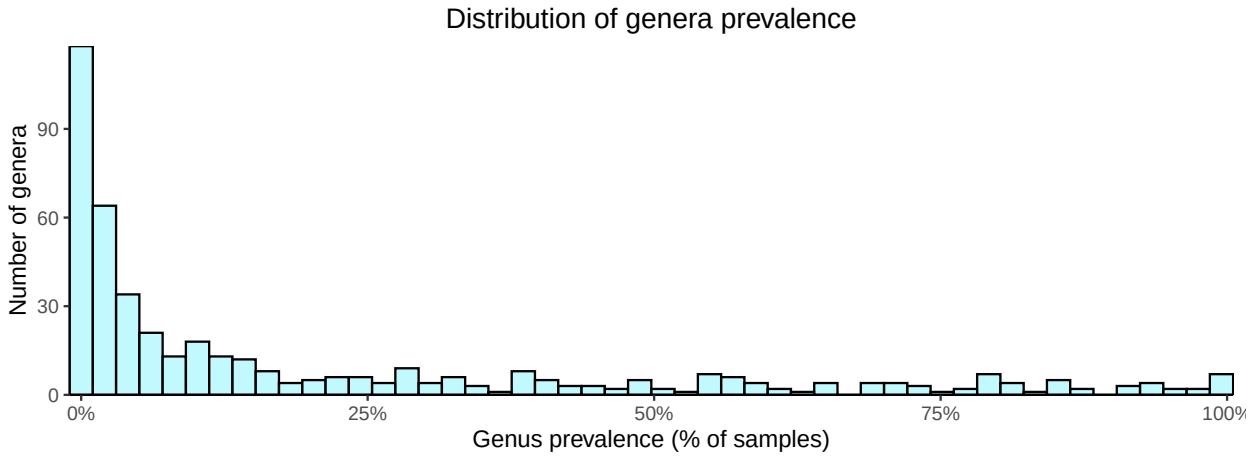


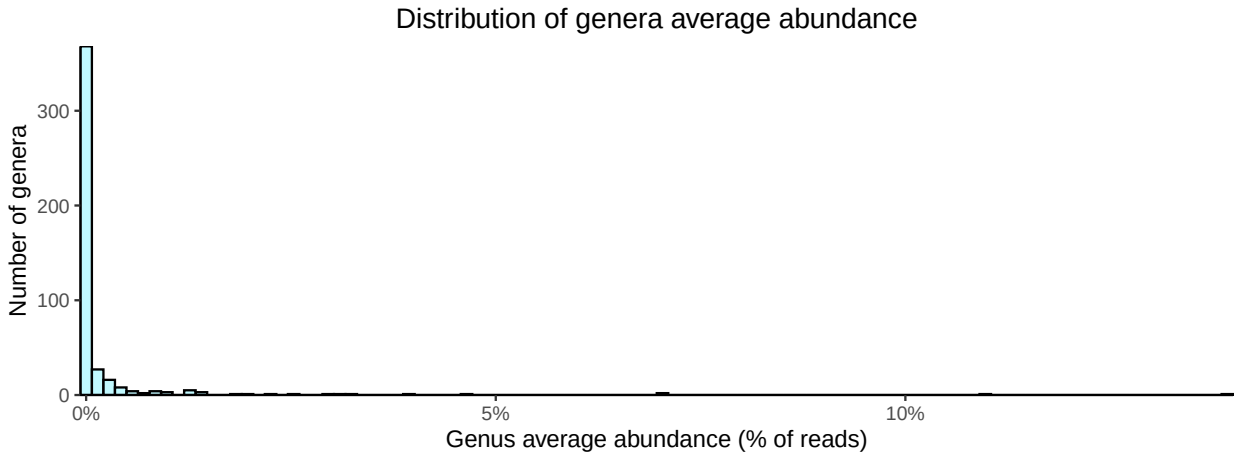
Genus-level taxonomy barplots



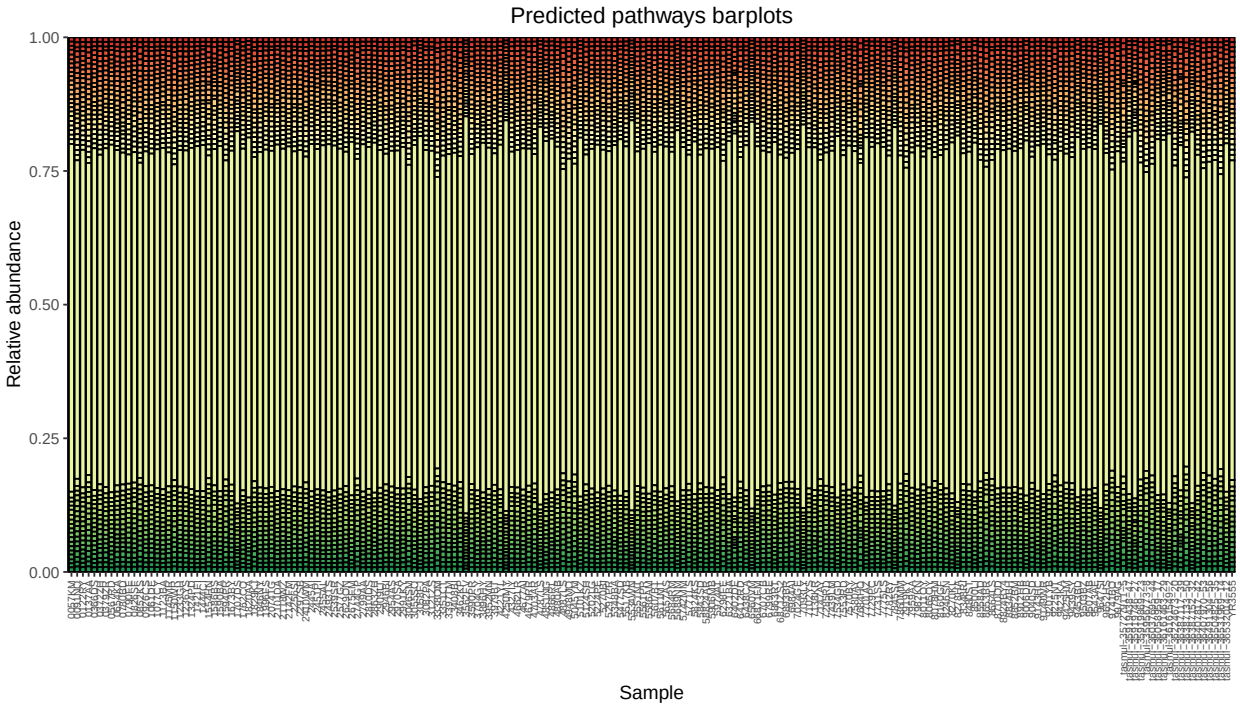
Genera prevalence and abundance

Below are the distributions of genera prevalence (% of samples in which each genus was detected) and average abundance (the abundance of each genus is averaged over all samples). These histograms will aid in defining thresholds to discard 'rare genera'.


Distribution of genera prevalence
Number of genera
90
60
30
0
0%
25%
50%
75%
100%
Genus prevalence (% of samples)


Distribution of genera average abundance
Number of genera
300
200
100
0
0%
5%
10%
Genus average abundance (% of reads)

Pathways predictions using PICRUSt2



Pathway

- 5−aminoimidazole ribonucleotide biosynthesis I
- 5−aminoimidazole ribonucleotide biosynthesis II
- adenosine deoxyribonucleotides de novo biosynthesis II
- adenosine ribonucleotides de novo biosynthesis
- Calvin−Benson−Bassham cycle
- CDP−diacylglycerol biosynthesis I
- CDP−diacylglycerol biosynthesis II
- chorismate biosynthesis from 3−dehydroquinate
- chorismate biosynthesis I
- cis−vaccenate biosynthesis
- glycogen biosynthesis I (from ADP−D−Glucose)
- glycogen degradation I (bacterial)
- glycolysis III (from glucose)
- gondoate biosynthesis (anaerobic)
- guanosine deoxyribonucleotides de novo biosynthesis II
- guanosine ribonucleotides de novo biosynthesis
- inosine−5'−phosphate biosynthesis I
- L−isoleucine biosynthesis I (from threonine)
- L−isoleucine biosynthesis II
- L−isoleucine biosynthesis III
- L−isoleucine biosynthesis IV
- L−lysine biosynthesis III
- L−lysine biosynthesis VI
- L−valine biosynthesis
- Others
- pentose phosphate pathway (non−oxidative branch)
- peptidoglycan biosynthesis I (meso−diaminopimelate containing)
- peptidoglycan biosynthesis III (mycobacteria)
- pyruvate fermentation to acetate and lactate II
- pyruvate fermentation to isobutanol (engineered)
- starch degradation V
- superpathway of 5−aminoimidazole ribonucleotide biosynthesis
- superpathway of adenosine nucleotides de novo biosynthesis I
- superpathway of adenosine nucleotides de novo biosynthesis II
- superpathway of aromatic amino acid biosynthesis
- superpathway of branched amino acid biosynthesis
- superpathway of L−isoleucine biosynthesis I
- superpathway of L−threonine biosynthesis
- superpathway of phospholipid biosynthesis I (bacteria)
- superpathway of pyrimidine nucleobases salvage
- UDP−N−acetylmuramoyl−pentapeptide biosynthesis I (meso−diaminopimelate containing)
- UDP−N−acetylmuramoyl−pentapeptide biosynthesis II (lysine−containing)
- UMP biosynthesis